Vincenzo Antonio Isoldi

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Research Interests

My research interests lie at the intersection of geometry, dynamics, and representation theory. I am currently learning about Anosov representations and (higher) Teichmüller theory, and I am fascinated by the interplay of analytic, algebraic, and differential techniques in answering geometric questions. On the more applied side, I enjoy thinking about the dynamics of interaction networks, their underlying geometry, and their ecological implications.

Education and Training

2026– **Ph.D. Student**
Max Planck Institute for Mathematics in the Sciences (MPI MiS)

Supervisor: Prof. Dr. Anna Wienhard.
Research group: Geometry, Groups, and Dynamics.

2026 **Pre-Doctoral Fellow**
Max Planck Institute for Mathematics in the Sciences (MPI MiS)
Max Planck Institute of Molecular Cell Biology and Genetics (MPI-CBG)

Supervisors: Profs. İrem Portakal (MPI MiS) and Türkü Özlüm Çelik (MPI-CBG).
Project on algebro-geometric, combinatorial, and game-theoretic aspects in the dynamics of interacting species.

2023–2025 **M.Sc. in Mathematics**
University of Trento

Final grade: 110/110 with honors. Final GPA: 30/30.
Thesis: "Schur Apolarity and the Dual Terracini Lemma for Flag Varieties: A Geometric and Representation-Theoretic Approach", supervised by Prof. Alessandra Bernardi and Dr. Stefano Canino.

2020–2023 **B.Sc. in Mathematics**
University of Milan

Final grade: 110/110.
Final seminar: "Plane Algebraic Curves: Local Study and Plücker Formulas".

2019–2020 **B.Sc. in Chemical Engineering**
Polytechnic University of Milan

GPA: 28.75/30. Mathematical coursework GPA: 30/30.
Interrupted to transition to B.Sc. in Mathematics.

Publications and Preprints

1. **Secant varieties of flag varieties via Schur apolarity**, with A. Bernardi and S. Canino. arXiv: 2609.09301.
2. **Stable Coexistence in Ecologies and Games**, with T. O. Çelik, I. Portakal, and G. Zucal. arXiv: 2609.09286.
3. **Neural Point-Forms**, with J. Hume, E. S. Ivshina, K. Maggs, P. Misof, and B. Trentini. arXiv: 2605.15524.

Honors, Awards, and Fellowships

2026 **Pearcy/Heep Graduate Mathematics Fellowship**, Texas A&M University
Awarded a one-time $2,500 fellowship in addition to a fully funded PhD offer. Declined.

2026 **Pre-Doctoral Fellowship**, Max Planck Institute for Mathematics in the Sciences
3-month fellowship. Amount: €1,650 per month.

2026 **Pre-Doctoral Fellowship**, Max Planck Institute of Molecular Cell Biology and Genetics
3-month fellowship. Amount: €1,753 per month.

2018 **Scholarship for the Scuola di Orientamento Universitario**, SNS, Sant'Anna, and IUSS
Selected among the top 230 Italian students.

2016 **WEP Scholarship**, WEP Italy
Awarded €1,000 for academic achievement, extracurricular involvement, and personal merit.

2016 **ITACA Scholarship**, INPS

Awarded €7,650 for a six-month study-abroad period during the academic year. 1st place nationally.

2013 **XXIII Olympiad of Logical and Mathematical Games**, Gioiamathesis
2nd place.

Teaching Experience

2024–2025 **Teaching Assistant**
University of Trento

Mathematical Analysis 3, Department of Physics, Prof. V. Agostiniani, A.Y. 2025/2026. Co-wrote weekly exercise sheets and solutions with A. Breggia.
Geometry and Linear Algebra, Department of Information Engineering and Computer Science, Prof. A. Bernardi, A.Y. 2025/2026, taught in English. Co-wrote the MATLAB lab sessions with F. G. De Vitis.
Geometry A, Department of Mathematics, Prof. E. L. Solá Conde, A.Y. 2024/2025.
Geometry and Linear Algebra, Department of Information Engineering and Computer Science, Prof. E. Postinghel, A.Y. 2024/2025.
Geometry and Linear Algebra, Department of Information Engineering and Computer Science, Profs. G. Occhetta and E. Postinghel, A.Y. 2024/2025, taught in English.
Mathematics and Statistics 1, Department of Biology, Profs. A. Bernardi and E. Ballico, A.Y. 2024/2025, 1:1.

Talks and Posters

Nov 2026 **Stable coexistence in ecologies and games**, CSBD Math Seminar, MPI-CBG (upcoming).

Nov 2026 **Stable coexistence in ecologies and games**, Seminar on Nonlinear Algebra, MPI MiS.

Mar 2026 **Flag varieties: from minimal orbits to a dual Terracini lemma**, TensorDec Lab Seminars, University of Trento.

Feb 2026 **Impossible ecologies via Grassmannians**, Coffee Chalk Talks, MPI-CBG.

Feb 2026 **Secant varieties of flag varieties**, PhD Selection Week, MPI-CBG. Poster.

Feb 2026 **Tensors and their geometry**, Mathematical Structures and Applications Group Seminars, MPI-CBG. Slides.

Jul 2025 **Geometric constrained variational calculus**, Seminar for Differential Geometric Methods, University of Trento.

Jul 2025 **Blow-up of $\mathbb{P}^2$ and minimal model program**, Seminar for Algebraic Geometry II, University of Trento.

Oct 2024 **Noether's normalization and homotopy continuation**, Seminar for Computational Algebra, University of Trento. Slides.

Sep 2024 **The algebraic geometry of tensor decomposition**, Baby Steps Beyond the Horizon, Będlewo. Slides.

Jun 2024 **Affine toric geometry**, Seminar for Toric Geometry, University of Trento.

Feb 2024 **Pattern formation in a model of acute inflammation**, Seminar for Mathematical Biology, University of Trento. Slides.

Conferences, Workshops, and Schools

Dic 2026 **AI4Math Workshop**, MPI MiS (upcoming)

Sep 2026 **Quasi-flats of intermediate dimension in higher rank**, MPI MiS

Jun 2026 **Workshop on Infinite-Dimensional Geometry**, MPI MiS

May 2026 **Character varieties under the lens of Hodge theory**, MPI MiS

Apr 2026 **AI in Math**, MPI MiS

Mar 2026 **GROW@Dresden 2026**, TU Dresden

Feb 2026 **3rd MPI (Dresden + Leipzig) Day**, MPI MiS

Jan 2026 **Algebraic Curves in January**, MPI MiS

Oct 2025 **Masterclass: Tensors and their Applications**, University of Trento

Aug 2025 **Scuola Matematica Interuniversitaria**, University of Perugia
Algebraic Geometry, Prof. Brian Harbourne, Grade: A+.
Complex Analysis, Prof. Chiara de Fabritiis, Grade: A.

Jul 2025 **London Geometry and Machine Learning**, Imperial College London
Selected participant, mentored by Dr. Kelly Maggs.

Jan 2025 **Apolarity in Varsavia**, Institute of Mathematics of the Polish Academy of Sciences
Invited by Prof. Jarosław Buczyński.
Co-authored workshop notes with O. R. Fité under the supervision of Dr. S. Canino.

Nov 2024 **Masterclass: Tensor Decompositions in Multi-Omics**, University of Trento

Jul 2024 **9th European Congress of Mathematics**, Seville

Oct 2023 **Masterclass: Algebraic Vision and Multifocal Tensors**, University of Trento

Service and Outreach

Oct 2026– **Teichmüller Theory Reading Group**, MPI MiS
Co-organized with Laura Lankers.

Mar 2026 **Il giro del mondo in 80 Prodotti**, Trento
Co-organized interactive booths on the history of multiplication for Pi Day.

Oct 2025 **TensorDay 2025**, University of Trento
Co-organized with the TensorDec Laboratory.

Apr 2025 **Fuori dall'ombra: Panchine Rosse**, University of Trento
Co-organized the *Matilda effect* section.

Dec 2024 **TensorDay 2024**, University of Trento
Co-organized with the TensorDec Laboratory.

Skills

Languages. Italian: native. English: fluent (TOEFL iBT 111/120).

Digital skills. Macaulay2, SageMath, C, MATLAB, LaTeX, Java.